

Three-dimensional Stress-relief Displacements from Blind-hole Drilling: a Parametric Description

by Ismael Rumzan and Douglas R. Schmitt

ABSTRACT—Optically-based interferometric techniques are finding increased application to the quantitative determination of near surface stress states. Unlike the standardized strain-gage method of hole drilling, however, some optical methods are sensitive to all three components of the displacement field produced by drilling of the stress-relieving hole. Analysis of the resulting fringe patterns necessitates a full knowledge of such motions. Here, direct formulae, which relate stress-relief displacements to radial position and azimuth, relative hole dimensions, residual or applied stress, and Poisson's ratio, are constructed from an extensive series of finite element calculations. The final formulae are derived from a large set of trial formulae that best describe the displacements according to a statistical regression analysis. The formulae are generally valid for hole depth/diameter ratios from 0.5 to 4.0, for Poisson's ratios from 0.05 to 0.45, and over radial distances from the hole axis from 2 to 20 times the hole radius, although these validity ranges can vary with hole depth. The equations are compared to an existing strain-gage hole-drilling standard and are used to forward model a speckle interferometer fringe pattern recording stress-relief displacements in an acrylic block.

KEY WORDS—Residual stress, blind-hole, stress-relief, ESPI, finite element

Introduction

The hole-drilling stress-relief method is one of the most popular quantitative stress measurement techniques. In application, local stress magnitudes and directions are estimated by measuring the surface deformation induced by the drilling of a small stress-relieving hole into the object of study. These deformations are mostly obtained using special rosette arrangements of strain gages and are recognized in an engineering standard.¹ A large amount of literature has evolved to deal with stress determination under uniform² and non-uniform^{3,4} near surface stress states, and with nonelastic deformation.⁵

In recent years there have been numerous attempts to employ optical interferometric methods to obtain this information by employing holography,^{6–11} moiré and Fizeau interferometry,¹² and correlative electronic speckle interferometry.^{13–19} These methods are potentially advantageous for a number of reasons including: (1) the object's surface needs little or no preparation; (2) time consuming,

highly accurate alignment of the drill relative to a strain-gage rosette is not necessary; (3) the fringe patterns over the object's surface are information rich and could provide displacements or strains over their full-field view, as suggested in an example fringe pattern²⁰ (Fig. 1(a)). There are a number of corresponding costs to optical methods, however. First, the sample and optical platform need to be mechanically stable during the test with relative motions less than only a small portion of the wavelength (λ) of the coherent light employed. Second, the very unwrapped nature of the fringe pattern (see, for example, Fig. 1(a)) must be properly accounted for. Finally, in experimental geometries that do not illuminate the object using plane waves the fringe patterns are often sensitive to the full three-dimensional deformation field including surface normal displacements in contrast to the two-dimensional in-plane strains of the popular strain-gage hole-drilling method.

This last issue is perhaps the greatest impediment to the application of optical methods because the relationship between the full-field stress-relief displacements and the applied stress is not well known. This makes problematic the inversion of the fringe patterns, which are directly related to the displacements, for stress magnitudes and directions. The magnitude and style of the displacement field, especially for out-of-plane displacements, depends not only on the elastic properties and the radius of the stress-relieving hole, but also on the depth of the blind hole.²¹ Unfortunately, there are no closed-form analytic solutions currently available to address this general problem. A closed-form solution is desirable as, aside from portability, it can readily be incorporated into recently developed fringe pattern inversion methodologies.²²

A number of workers have made attempts to overcome this situation even for the simpler two-dimensional strain-gage technique, where the general form of the equations describing the azimuthal dependence (Fig. 2) of radial stress-relief strain, ϵ_r , on a uniaxial stress state, σ_{xx} , is¹

$$\epsilon_r = [A + B \cos(2\theta)]\sigma_{xx}, \quad (1)$$

where the coefficients A and B are calibration constants. Presently, only the uniaxial case is considered for purposes of clarity in illustration. For the theoretical case of a through hole in a stressed infinite plate²³ the constants are

$$A = -\frac{1-\nu}{2E} \frac{1}{\rho^2} \quad (2)$$
$$B = -\frac{1+\nu}{2E} \left[\frac{4}{1+\nu} \frac{1}{\rho^2} - 3 \frac{1}{\rho^4} \right].$$

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Original manuscript submitted: November 1, 2001.

Final manuscript received: June 17, 2002.

where $\rho = r/r_o$ is the dimensionless radius (Fig. 2). This simple formula, however, illustrates explicitly the harmonic dependence of the deformation on the azimuth and on the elastic moduli and hole dimensions. In the more realistic case of a blind hole, the situation is not so straightforward and the geometry must be considered. Consequently the coefficients employed in practice implicitly depend on the experimental geometry (i.e., gage area and hole depth) and the elastic properties. Such coefficients are either obtained experimentally² or by numerical analysis.^{24,25}

The optical interferometric case provides more of a challenge because of the possibility that three-dimensional information is required. Because the problem no longer relies on highly specific experimental geometries, such as the gage area, it is more convenient to work in non-dimensional parameters to make the solution more readily available. In particular, the “basis shape” of a given stress-relief displacement field will scale directly for constant hole dimensions and the magnitude of the real displacements are linearly related to Young’s modulus, E . As such, the dimensionless radius is $\rho = r/r_o$ and the hole depth may be represented as $\delta = h/(2r_o)$ relative to its diameter. Similarly, a normalized displacement component $\bar{u}_i = u_i/r_o$ where the subscript i refers to any one of the three co-ordinates, ρ , θ , or z . The most general series of formulae describing the three normalized stress-relief displacements (Fig. 2) at constant radius ρ at a given point $P(\rho, \theta)$ would take the form

$$\begin{aligned}\bar{u}_r &= [A_r + B_r \cos(2\theta)] S_{yy} \\ \bar{u}_\theta &= [B_\theta \sin(2\theta)] S_{yy} \\ \bar{u}_z &= [A_z + B_z \cos(2\theta)] S_{yy},\end{aligned}\quad (3)$$

where $S_{yy} = \sigma_{xx}/E$ is the normalized stress. Note that the constant A_θ must be equal to zero because the symmetry of the problem requires that $\bar{u}_\theta$ vanish at $\theta = 0, \pi/2, \pi$ and $3\pi/2$. The two normalized coefficients, A_i and B_i , are now implicitly functions of ρ , δ , and Poisson’s ratio ν ; a number of specific cases have been provided in the literature. Furgiuele *et al.*²⁶ have given coefficients for the stress-relief displacements at the edge of the stress-relieving hole ($\rho = 1$) first illustrating the strong influence of both ν and δ . Lin *et al.*⁹ have given the coefficients for the vertical stress-relief displacements for ranges of $1.2 \geq \delta \geq 0.8$, and $0.35 \geq \nu \geq 0.25$ for ρ from 1.8 to 3.8. They have also indicated the care that must be taken with boundary conditions in the development of the finite element model used as a basis. Makino *et al.*¹¹ have provided graphs of surface in-plane coefficients dependent on ρ and δ and effective over the range $0.35 \geq \nu \geq 0.25$; these were used to successfully provide stresses from holographic fringe patterns. Despite this progress, some optical experimental geometries require knowledge of the full three-dimensional set of stress-relief displacements.²² Also, there is some need to carry out stress determination in geophysical or polymeric materials whose Poisson’s ratios principally differ from those for metallic alloys.

To overcome the above difficulties, an extensive series of finite element calculations for stress-relief displacements were carried out over wide ranges of both Poisson’s ratio ν and the dimensionless hole depth δ . The resulting finite element displacements were then optimally fitted by testing large numbers of trial formulae to yield a series of analytic expressions that directly give, in non-dimensional form, the

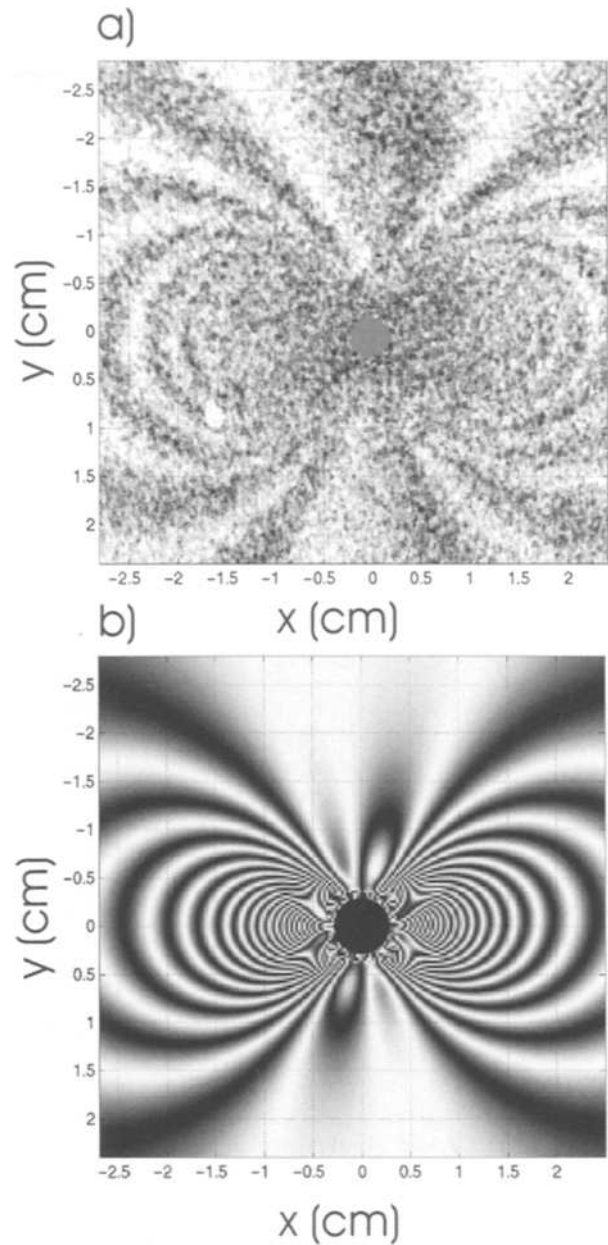


Fig. 1—Comparison of (a) observed over acrylic and (b) forward modeled stress-relief fringe patterns calculated using derived analytic expressions for an applied uniaxial stress state $\sigma_{xx} = 5.62$ MPa, $\sigma_{yy} = 0$, $\tau_{xy} = 0$, for hole dimensions $\delta = 2.38$, $r_o = 0.2855$ cm, in a material of elastic properties $E = 3.1$ GPa and $\nu = 0.4$. The origin is centered at the intersection of the stress-relief hole and the plane of the sample surface. The source positions²² are: S_1 , $x = -91.02$ mm, $y = 35.76$ mm, $z = 39.36$ mm; S_2 , $x = 79.39$ mm, $y = 46.20$ mm, and $z = 44.05$ mm

displacements in terms of ρ , δ , ν and stress. While this procedure is not necessarily elegant and the essential physics of the problem is obscured, it does have the advantage of giving portable closed-form solutions to the complex three-dimensional stress-relief field. The equations are tested by comparison to existing strain-gage standards and by forward modeling of an observed fringe pattern obtained in the laboratory.

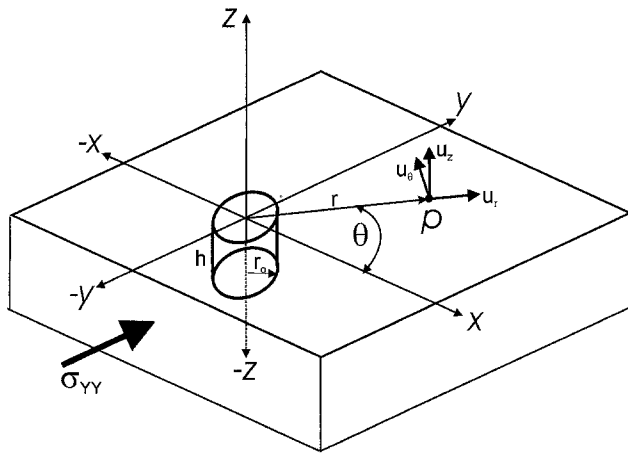


Fig. 2—Geometry of a blind hole of depth h and radius r_o within a block subject to uniaxial normal compression σ_{yy} . A positive sense of induced stress-relief displacements and of stress (compression reckoned positive) is employed in this study

Finite Element Modeling

Finite element modeling was employed to calculate the stress-relief displacements in order to overcome the difficulties associated with determination of the three-dimensional stress-relief displacements induced by the drilling of a blind hole. As in earlier studies²⁷ only the case of uniaxial compression was calculated. The set of basis displacements resulting from these calculations readily allows for the determination of a more complex biaxial stress state by appropriate rotation and superpositioning of the displacements,²⁸ but it is not presented here. For such a case, the model mesh may be maximized by taking advantage of the symmetry of the problem with respect to the vertical planes perpendicular and parallel to the applied compression σ_{yy} that intersect along the axis of the stress-relief hole. This allows the region of calculation to be reduced to a single quadrant around the stress-relief hole as long as appropriate no-motion boundary conditions are applied at the edges of the block.¹⁰ The finite element mesh (Fig. 3) consisted of 6600 eight-node brick elements over a block with dimensions $15 \times 15 \times 6 \text{ cm}^3$ in the x -, y -, and z -directions, respectively. A constant hole radius, $r_o = 2.5 \text{ mm}$, was employed with depths ranging from 2.5 to 20 mm providing a range of $4.0 \geq \delta \geq 0.5$. The calculations were carried out using a commercial finite element analysis (FEA) code with a normalized compression $S_{yy} = 1.935 \times 10^{-4}$ with values of $\nu = 0.05, 0.25$, and 0.45 . It must be noted that compression is reckoned to be positive in this contribution; the use of the tension positive convention requires only that the polarity of the coefficients developed be reversed.

The finite element method does not yield stress-relief displacements directly. They may only be obtained by superposition as the difference in the displacements of the block containing the stress-relief hole less those for the same block with no stress-relief hole subject to the same stress and boundary conditions.²⁹ Some care must be exercised in the boundary conditions⁹ and the conditions employed by Li²⁷ that appear to best agree with laboratory experiments are retained. Symmetric boundary conditions (i.e., in-plane rotation and out-of-plane translation disallowed) were applied to the x - z and y - z sides containing the blind-hole axis and the bottom x - y plane. The displacements of the opposite x - z (where the

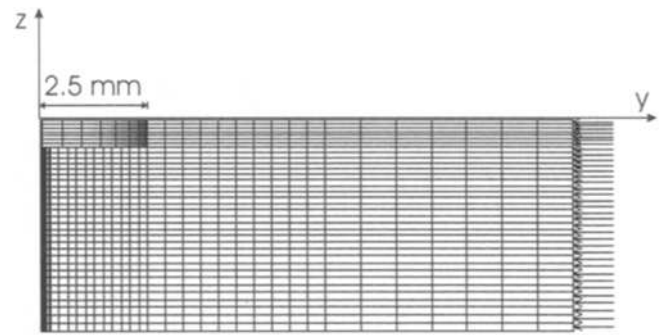


Fig. 3—A side view of the detailed view of a portion of the finite element mesh in the vicinity of the stress-relieving hole of $r_o = 2.5 \text{ mm}$ and $h = 0.5 \text{ mm}$

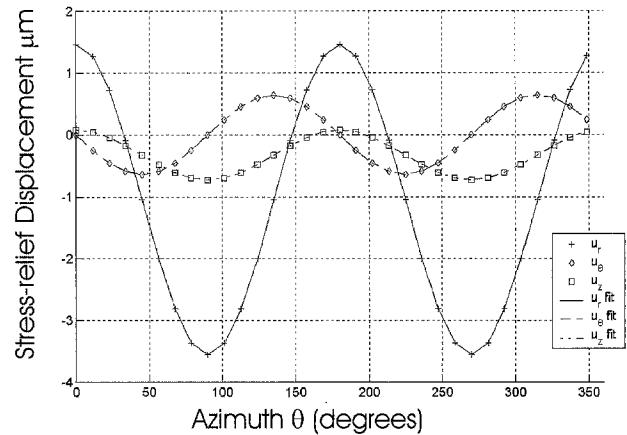


Fig. 4—Harmonic dependence of stress-relief displacements with azimuth θ at $\rho = 2.0$ for the case of $\nu = 0.25$, $\delta = 2$ ($r_o = 2.5 \text{ mm}$), and normalized stress $S_{yy} = 1.935 \times 10^{-4}$

normal stress σ_{yy} is applied) and y - z sides, the top x - y surface, and the hole were unconstrained. The success of this set of boundary conditions was monitored by examining the differences in the displacements calculated for the block relative to the block containing a hole; this difference became effectively negligible past $\rho \sim 20$. An example of the harmonic dependence of the stress-relief displacements at constant radius is given in Fig. 4.

For constant stress-relief hole dimensions and applied stress, the variation in the displacement magnitudes over the range of ν from 0.05 to 0.45 is substantial (Fig. 5). As might be expected, u_r increases only weakly on ν whereas, owing to the Poisson effect, both u_θ and u_z are highly sensitive, increasing by large factors over the range of ν studied. Optical interferometric methods are often sensitive to the out-of-plane displacements and hence the analysis of resulting fringe patterns must more carefully incorporate the influence of ν than do more standard strain-gage techniques in which radial deformation is recorded.

The hole depth, δ , also has a large influence on the stress-relief displacement field (Fig. 6). All displacement components, including u_r , increase rapidly with δ . The influence of the normalized hole depth δ is particularly striking on u_z ; the magnitude of this stress-relief displacement continues to grow with the hole depth although the rate of change of the displacements with larger δ rapidly diminishes.²¹ These results further demonstrate how carefully the three-dimensional

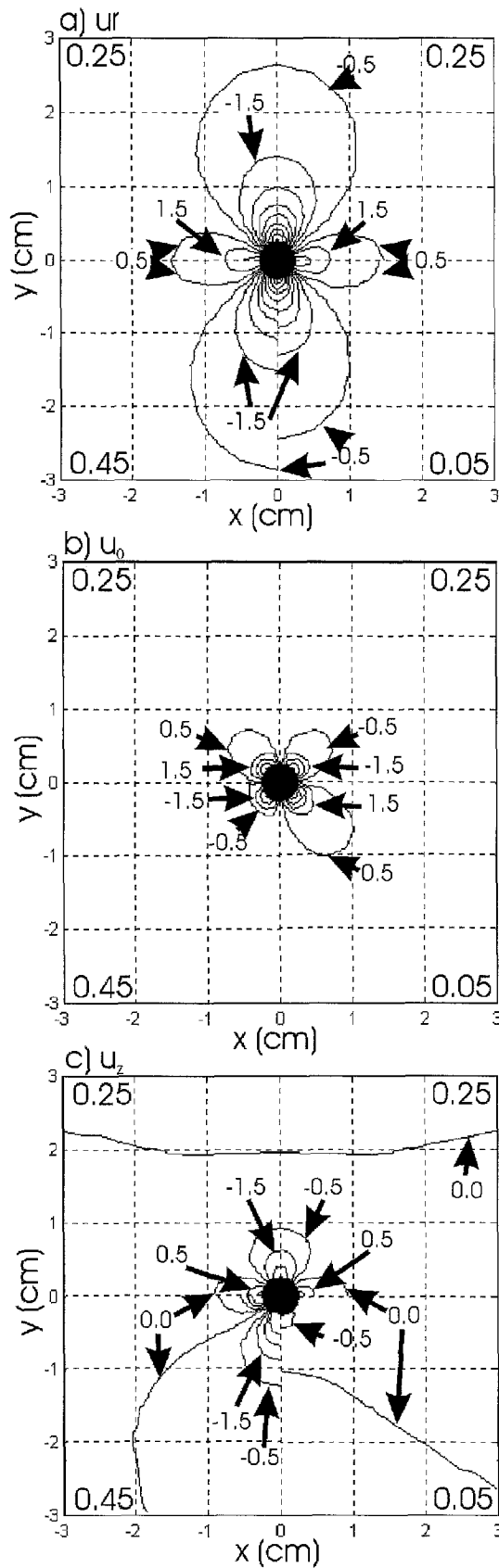


Fig. 5—Influence of Poisson's ratio on (a) u_r , (b) u_θ , and (c) u_z for the case with $E = 3.1$ GPa, $\sigma_{yy} = 6.0$ MPa for a constant $\delta = 2$ ($r_o = 2.5$ mm, $h = 10$ mm). The contours (in μm) within both upper quadrants, the left bottom quadrant, and the right bottom quadrant show the displacement magnitudes for values of Poisson's ratio of 0.25, 0.45, and 0.05, respectively

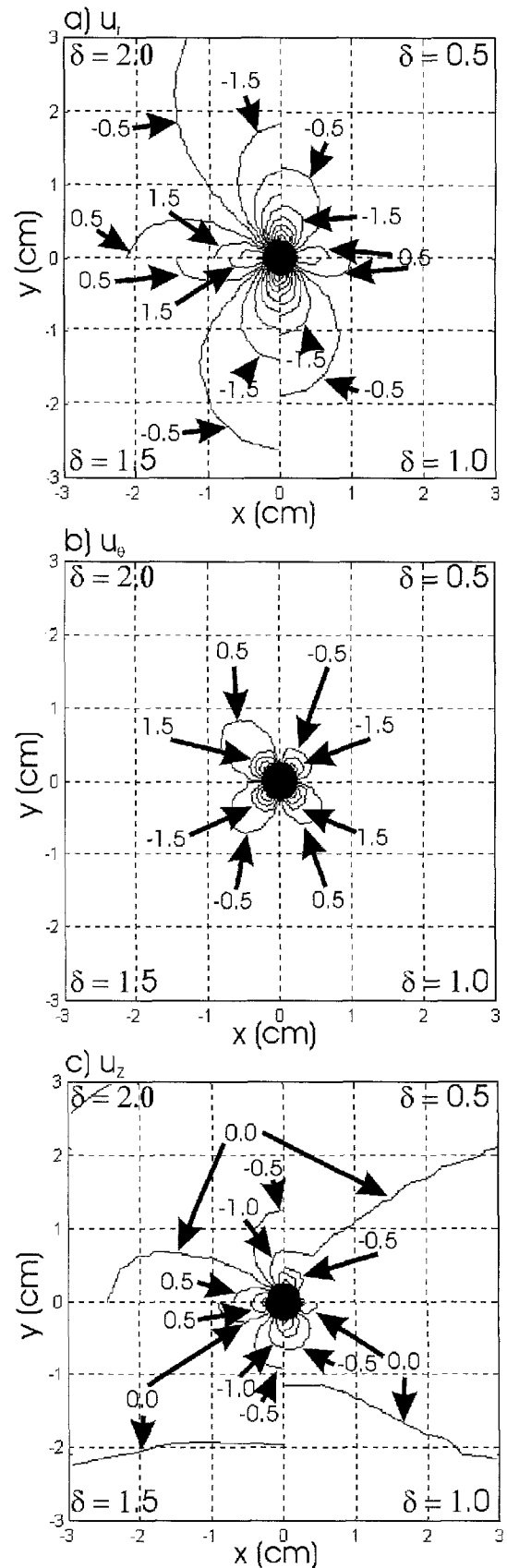


Fig. 6—Influence of the hole depth ratio $\delta = h/2r_o$ ($r_o = 2.5$ mm) on (a) u_r , (b) u_θ , and (c) u_z for the case with $E = 3.1$ GPa, $\sigma_{yy} = 6.0$ MPa for a constant $\nu = 0.25$ and for values of $\delta = 0.5, 1.0, 2.0$, and 4.0 respectively by quadrants in a clockwise direction. The contours are in μm

components of the displacement field must be understood in order to properly employ interferometric fringe patterns in finding residual stress.

In this study, numerical calculations are essentially a virtual substitute for an extensive series of calibration tests.² As such, the reliability of the calculations must be questioned. The quality of the finite element calculations was validated by comparing the displacements obtained around the circumference of the long cylindrical cavity to the classic analytic solution of Kirsch;²³ the FEA and analytic solutions differed by less than 2 percent.

Derivation of Parametric Coefficients

The primary goal of this contribution is to provide general formulae for the stress-relief displacements. A given displacement, however, depends in a complex fashion on azimuth θ , hole depth δ , radius ρ , and Poisson's ratio ν in addition to being scaled by the applied stress. The value of the stress may be ignored further as the problem is given here in dimensionless coordinates. An initial examination of eqs (3) indicates that for any given ν and δ the value of each of the coefficients at constant radius ρ may be found as

$$\begin{aligned} A_r(\nu, \delta)|_\rho &= \frac{\bar{u}_r(\nu, \delta, \theta = \pi/4)|_\rho}{S_{yy}} \\ B_r(\nu, \delta)|_\rho &= \frac{\bar{u}_r(\nu, \delta, \theta = 0)|_\rho - \bar{u}_r(\nu, \delta, \theta = \pi/4)|_\rho}{S_{yy}} \\ B_\theta(\nu, \delta)|_\rho &= \frac{\bar{u}_\theta(\nu, \delta, \theta = \pi/4)|_\rho}{S_{yy}} \\ A_z(\nu, \delta)|_\rho &= \frac{\bar{u}_z(\nu, \delta, \theta = \pi/4)|_\rho}{S_{yy}} \\ B_z(\nu, \delta)|_\rho &= \frac{\bar{u}_z(\nu, \delta, \theta = 0)|_\rho - \bar{u}_z(\nu, \delta, \theta = \pi/4)|_\rho}{S_{yy}}, \end{aligned} \quad (4)$$

where it is important to reiterate that the coefficients developed here are for the situation as represented in Fig. 2, i.e., for a uniaxial stress applied along the y -axis with θ measured counterclockwise from the positive x -axis. As a first step in obtaining a series of coefficients, eq (4) is applied to the results of all the finite element models to yield sets of Poisson's ratio and δ specific coefficients A_i and B_i at each radius.

The large number of coefficients that resulted were then further selectively grouped by constant δ yielding $A_i(\nu)|_{\rho, \delta}$ and $B_i(\nu)|_{\rho, \delta}$ ($i = r, \theta, z$) in order to examine in isolation the influence of Poisson's ratio. The coefficients were found to depend linearly or quadratically on ν according to

$$\begin{aligned} A_r(\delta, \rho, \nu) &= (1 + \nu)A_{r1} \\ B_r(\delta, \rho, \nu) &= B_{r1}\nu + B_{r2} \\ B_\theta(\delta, \rho, \nu) &= B_{\theta1}\nu^2 + B_{\theta2}\nu - B_{\theta3} \\ A_z(\delta, \rho, \nu) &= (1 + \nu)A_{z1} \\ B_z(\delta, \rho, \nu) &= B_{z1}\nu + B_{z2}, \end{aligned} \quad (5)$$

i.e., for constant ρ and δ a simple relationship between Poisson's ratio and the coefficients exists that may be described

by a lower set of coefficients which now depend only on ρ and δ . Equation (5) was then applied to each of the many possible combinations of ρ and δ to yield nine surfaces $A_{ij}(\rho, \delta)$ and $B_{ij}(\rho, \delta)$ ($i = r, \theta, z$ and $j = 1, 2, 3$ in relation to eq (5)).

Unfortunately, these surfaces are not simply described as illustrated by an example (Fig. 7) of the radial dependence of the three $B_{\theta i}$ coefficients for the case of $\delta = 2$. A least-squares regression testing of a large number of candidate formulae was employed³⁰ to overcome this limitation and this process resulted in the following formulae for $A_{ij}(\rho, \delta)$ and $B_{ij}(\rho, \delta)$:

$$\begin{aligned} A_{r1} &= \frac{A_{r1}^1 + A_{r1}^2 \ln(\rho) + A_{r1}^3 \ln(\delta) + A_{r1}^4 \ln(\rho)^2 + A_{r1}^5 \ln(\delta)^2 + A_{r1}^6 \ln(\rho) \ln(\delta)}{1 + A_{r1}^7 \ln(\rho) + A_{r1}^8 \ln(\delta) + A_{r1}^9 \ln(\rho)^2 + A_{r1}^{10} \ln(\delta)^2 + A_{r1}^{11} \ln(\rho) \ln(\delta)} \\ B_{r1} &= \frac{B_{r1}^1 + B_{r1}^2 \ln(\rho) + B_{r1}^3 \ln(\delta) + B_{r1}^4 \ln(\rho)^2 + B_{r1}^5 \ln(\delta)^2 + B_{r1}^6 \ln(\rho) \ln(\delta)}{1 + B_{r1}^7 \ln(\rho) + B_{r1}^8 \ln(\delta) + B_{r1}^9 \ln(\rho)^2 + B_{r1}^{10} \ln(\delta)^2 + B_{r1}^{11} \ln(\rho) \ln(\delta)} \\ B_{r2} &= \frac{B_{r2}^1 + B_{r2}^2 \ln(\rho) + B_{r2}^3 \ln(\delta) + B_{r2}^4 \ln(\rho)^2 + B_{r2}^5 \ln(\delta)^2 + B_{r2}^6 \ln(\rho) \ln(\delta)}{1 + B_{r2}^7 \ln(\rho) + B_{r2}^8 \ln(\delta) + B_{r2}^9 \ln(\rho)^2 + B_{r2}^{10} \ln(\delta)^2 + B_{r2}^{11} \ln(\rho) \ln(\delta)} \\ B_{\theta1} &= \frac{1}{B_{\theta1}^1 + B_{\theta1}^2 \rho^{1.5} + B_{\theta1}^3 \delta^{1.5} + B_{\theta1}^4 \exp(-\delta)} \\ B_{\theta2} &= B_{\theta2}^1 + \frac{B_{\theta2}^2}{\rho} + B_{\theta2}^3 \ln(\delta) + \frac{B_{\theta2}^4}{\rho^2} + B_{\theta2}^5 \ln(\delta)^2 \\ &\quad + B_{\theta2}^6 \frac{\ln(\delta)}{\rho} + \frac{B_{\theta2}^7}{\rho^3} + B_{\theta2}^8 \ln(\delta)^3 + \\ &\quad B_{\theta2}^9 \frac{\ln(\delta)^2}{\rho} + B_{\theta2}^{10} \frac{\ln(\delta)}{\rho^2} \end{aligned} \quad (6)$$

$$\begin{aligned} B_{\theta3} &= \frac{B_{\theta3}^1 + B_{\theta3}^2 \ln(\rho) + B_{\theta3}^3 \ln(\delta) + B_{\theta3}^4 \ln(\rho)^2 + B_{\theta3}^5 \ln(\delta)^2 + B_{\theta3}^6 \ln(\rho) \ln(\delta)}{1 + B_{\theta3}^7 \ln(\rho) + B_{\theta3}^8 \ln(\delta) + B_{\theta3}^9 \ln(\rho)^2 + B_{\theta3}^{10} \ln(\delta)^2 + B_{\theta3}^{11} \ln(\rho) \ln(\delta)} \\ A_{z1} &= \frac{A_{z1}^1 + A_{z1}^2 \ln(\rho) + A_{z1}^3 \ln(\delta) + A_{z1}^4 \ln(\rho)^2 + A_{z1}^5 \ln(\delta)^2 + A_{z1}^6 \ln(\rho) \ln(\delta)}{1 + A_{z1}^7 \ln(\rho) + A_{z1}^8 \ln(\delta) + A_{z1}^9 \ln(\rho)^2 + A_{z1}^{10} \ln(\delta)^2 + A_{z1}^{11} \ln(\rho) \ln(\delta)} \\ B_{z1} &= \frac{B_{z1}^1 + B_{z1}^2 \rho + B_{z1}^3 \rho^2 + B_{z1}^4 \rho^3 + B_{z1}^5 \delta + B_{z1}^6 \delta^2}{1 + B_{z1}^7 \rho + B_{z1}^8 \delta + B_{z1}^9 \delta^2 + B_{z1}^{10} \delta^3} \\ B_{z2} &= \frac{B_{z2}^1 + B_{z2}^2 \rho + B_{z2}^3 \ln(\delta) + B_{z2}^4 \ln(\rho)^2 + B_{z2}^5 \ln(\delta)^2 + B_{z2}^6 \ln(\rho) \ln(\delta)}{1 + B_{z2}^7 \rho + B_{z2}^8 \ln(\delta) + B_{z2}^9 \ln(\rho)^2 + B_{z2}^{10} \ln(\delta)^2 + B_{z2}^{11} \ln(\rho) \ln(\delta)}, \end{aligned}$$

where the subcoefficients A_{ij}^k and B_{ij}^k are constants presented in Table 1. The formulae in eq (6) are those that best fit the coefficient surfaces; in all cases the coefficient of correlation between the formulae and the surfaces was better than 0.997. In the immediate vicinity of the stress-relieving hole, the displacements were especially complex. Including these into the fitting procedure considerably lowered the quality of the regression. Similarly, at large radii the stress-relief displacements essentially vanished leaving only numerical noise. As such, the fitting procedures used to obtain eq (6) were generally restricted to the range $20 \geq \rho \geq 2$ although

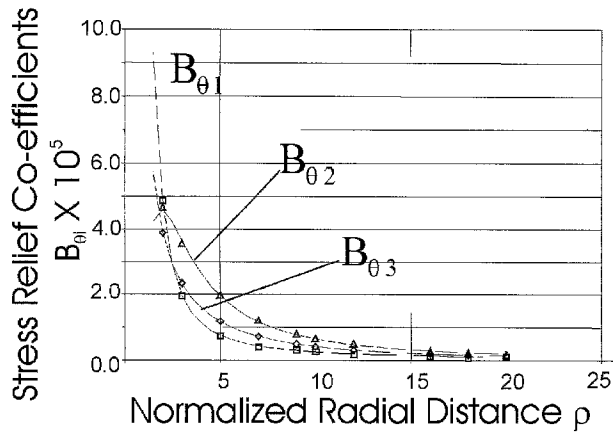


Fig. 7—Radial dependence of displacement coefficients $B_{\theta 1}$ (open squares), $B_{\theta 2}$ (open triangles), and $B_{\theta 3}$ (open diamonds) for the case $\delta = 2$. The symbols are the coefficients directly calculated from the original finite element calculations and the lines are the corresponding analytic description of eq (6)

this depends to some degree on δ and the reader must take note of the validity ranges given in Table 1. From a practical perspective, loss of information very near the hole is not crucial; this region is often subject to anelastic deformation that is not accounted for in this study. Furthermore, the speckle interferometric method does not have sufficient spatial resolution to be useful very near the stress-relieving hole (Fig. 1(a)).

Briefly, to apply these formulae the coefficients $A_{ij}(\rho)$ and $B_{ij}(\rho)$ are first obtained for the particular hole geometry δ of the specific problem using eq (6). The influence of Poisson's ratio is then included in the next level of coefficients $A_i(\rho)$ and $B_i(\rho)$ using eq (5). These resulting coefficients $A_i(\rho)$ and $B_i(\rho)$ are then employed with the normalized stress S_{yy} in eq (3) to give the dimensionless stress-relief displacements. These may then be converted to true displacements by $u_i = r_o \bar{u}_i$. An example of carrying out this calculation for a variety of simple cases is compared against the original finite element values in Fig. 8.

Comparisons to Strain-gage Hole Drilling

The strain-gage hole-drilling technique relies on the measurement of the stress-relief radial strains. As noted in eq (1) stress determination will require an appropriate set of coefficients that takes into account the particular strain-gage and stress-relief hole geometries. These coefficients are provided in the existing American Society for Testing and Materials (ASTM) standard,¹ a subset of which is reproduced in Table 2. These standards were also derived³¹ from finite element modeling and agree to better than 5 percent with earlier published experimental results. Consequently, the standards provide a useful reference against which the present results may be compared.

To make the comparison, the stress-relief displacements of eq (3) must be converted into the appropriate strains accounting for the placement of the strain gages relative to the stress-relieving hole. Calculations were carried out for the specific geometry of the strain gages given in Fig. 1 of Ref. 1 ($D = 5.13$ mm) with a variety of values of the ratio r_o/r_m , where r_m is the radius passing through the center of the strain

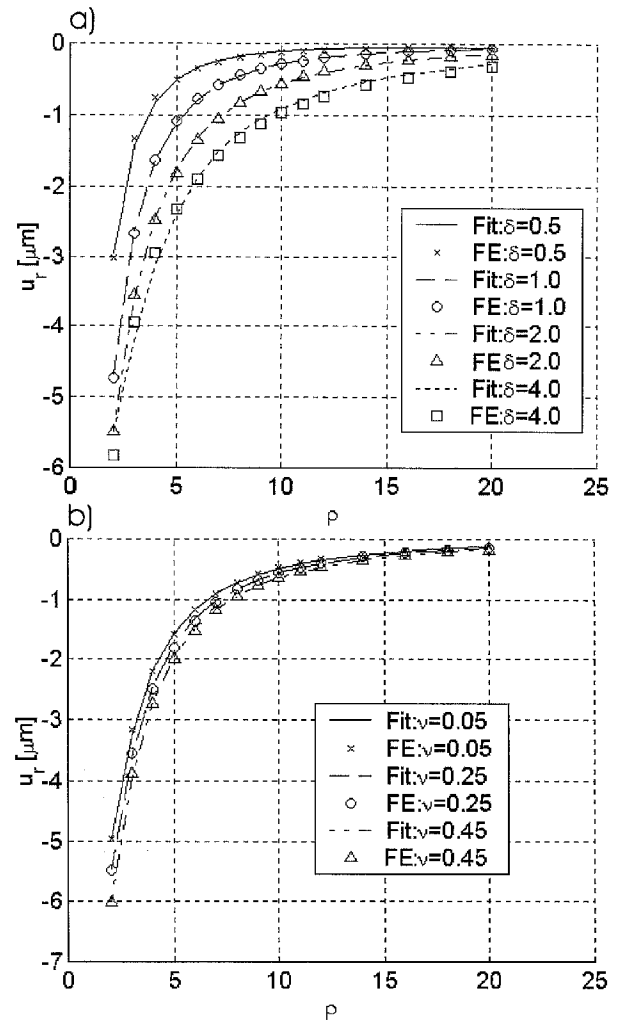


Fig. 8—Comparison between displacements produced by finite element calculations and the corresponding parametric fits under normalized stress $S_{yy} = 1935 \times 10^{-3}$ for: (a) u_r versus ρ at $\theta = \pi/2$ and $\nu = 0.25$ for a variety of values of δ ($r_o = 0.25$ cm); (b) u_z versus ρ at $\theta = \pi/2$ and $\delta = 2$ ($r_o = 0.25$ cm) for a variety of values of Poisson's ratio

gages (Fig. 9). This was accomplished here by first calculating the stress-relief displacement field for unit normalized stress, then differentiating these fields to yield the radial ϵ_r and azimuthal ϵ_θ strains according to

$$\begin{aligned} \epsilon_r(r, \theta) &= \frac{r_o \partial \bar{u}_r}{\partial r} \\ \epsilon_\theta(r, \theta) &= \frac{r_o \bar{u}_r}{r} + \frac{r_o \partial \bar{u}_\theta}{r \partial \theta} \end{aligned} \quad (7)$$

from which the gage longitudinal strain $\epsilon_L = \epsilon_r \cos(\alpha) + \epsilon_\theta \sin(\alpha)$ is calculated. This strain is then numerically integrated over the area of the gage of length l , width $2s$, and smallest radial distance R_1 (Fig. 9), in a manner similar to that used by Kabiri:²⁹

$$[\epsilon]_r = \frac{1}{2ls} \int_{R_1}^{R_1+l} \int_{-s}^s \epsilon_L dy dx, \quad (8)$$

TABLE 1—COEFFICIENTS FOR PARAMETRIC FORMULAE (DIMENSIONLESS)

i	A_r^i	B_{r1}^i	B_{r2}^i	$B_{\theta 1}^i$ ($\times 10^{-5}$) [†]	$B_{\theta 2}^i$	$B_{\theta 3}^i$	A_z^i	B_{z1}^i	B_{z2}^i
1	417453	454999	788230	-1.01806	-12891	558155	114751	37208	70156
2	-259172	-244193	-399590	0.12903	268001	-304060	-89755	-8224.9	-4870.0
3	202840	186758	480564	0.49225	-39625	288877	49308	359.53	1498.8
4	41329	35977	50974	1.19169	2319321	42756	16858	-6.2243	71.022
5	23609	19836	49301	—	4802.8	39144	4564.2	32434	-32843
6	-63933	-48725	-142327	—	868229	-83120	-19017	-3657.5	3116.8
7	-0.10124	-0.62163	-0.82025	—	-2932972	0.09405	-1.07633	0.03724	-0.11721
8	0.38363	0.82249	-0.53539	—	6158.4	-0.03836	1.27917	-1.75988	0.99042
9	0.31838	0.49803	1.04032	—	23936	0.51267	0.91871	0.88620	0.01565
10	0.28399	0.61413	0.19595	—	-728401	0.19162	0.85083	-0.12668	-0.43376
11	-0.01177	-0.09131	-0.59772	—	—	0.05177	-1.51063	—	-0.11865

Coefficients describing r and θ displacements are valid over the ranges $0.45 \geq \nu \geq 0.05$, $20 \geq \rho \geq 2$, $4.0 \geq \delta \geq 0.5$.

Coefficients describing z displacements are valid over the ranges of $0.45 \geq \nu \geq 0.05$ but only $5 \geq \rho \geq 2$ for $\delta = 0.5$, $9 \geq \rho \geq 2$ for $\delta = 1.0$, $20 \geq \rho \geq 2$ for $\delta = 2.0$, and $28 \geq \rho \geq 2$ for $\delta = 4.0$.

[†]Multiply table value by 10^{-5} in order to obtain proper value to use in eqs (6); for example, the value of $B_{\theta 1}^1$ that needs to be used is -1.01806×10^{-5}

TABLE 2—COMPARISON WITH EXISTING STANDARDS FOR THE STRAIN-GAGE HOLE-DRILLING TECHNIQUE FOR A CONSTANT HOLE DEPTH TO ROSETTE-GAGE DIAMETER $H/D = 0.4$ AND FOR $\nu = 0.3$

r_o/r_m	δ	Gage Radius R_1	Gage Radius R_2	Coefficients Published in ASTM Standard		Coefficients Calculated Using Parametric Solution	
				$\bar{a}$	$\bar{b}$	$\bar{a}$	$\bar{b}$
0.3	1.333	2.33	4.33	0.1117	0.288	0.1124	0.2741
0.35	1.143	2.00	3.71	0.150	0.376	0.1485	0.3807
0.40	1.000	1.75	3.25	0.190	0.466	0.1857	0.4872
0.45	0.889	1.56	2.89	0.236	0.558	0.2222	0.5831
0.50	0.800	1.40	2.60	0.285	0.648	0.2563	0.6593

where $[\epsilon_r]$ is the “effective” radial strain from which the coefficients in eq (1) are determined by carrying out the calculations at $\theta = 0$ and $\pi/4$. It is worth noting that $[\epsilon_r]$ could also be determined by more directly employing the stress-relief displacements.³² Equation (8) does not incorporate any transverse gage sensitivity. A series of calculations, in which small influences of the vertical strains are included, have shown that this component of the deformation has negligible influence on the final coefficients and is subsequently ignored.

The values of the coefficients for this case are close to, but not exactly the same as, those given in the standard.¹ On average, the present coefficients differ from the standard by slightly less than 4 percent although the variation appears to increase with the ratio r_o/r_m . This may be partly due to the fact that for $r_o/r_m >$, R_1 starts to fall outside the range of validity of our equations (i.e., $R_1/r_o < 2$) as indicated in Table 2. Hence, one would need to be careful in using these values in this range.

Comparison to Interferometric Pattern

The primary motivation for the present study is to develop a methodology for calculating the entire set of stress-relief displacements for use with optical interferometric measurements that can be sensitive to the entire displacement field. In particular, correlative speckle interferometry detects the displacements that occur in the time interval between reference and measurement images. The example observed fringe pat-

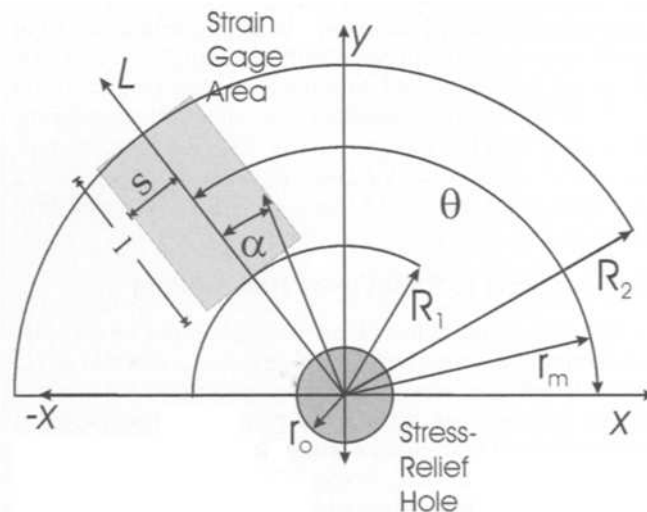


Fig. 9—Geometry of strain gage for determination of coefficients. The gage is sensitive to strains ϵ_L parallel to gage axis L at angle θ . Gage dimensions are length l and width $2s$ with the gage lying between radii R_1 and R_2

tern (Fig. 1(a)) was created by drilling a blind hole ($\delta = 2.38$, $r_o = 0.2855$ cm) into an annealed acrylic block uniaxially loaded with a compressive stress $\sigma_{xx} = 5.62$ MPa using the experimental procedure outlined in Ref. 10.

The object is illuminated by two source points that expand both parts of a split coherent laser beam. More details of the experimental geometry may be found in Jones and Wykes.³³ The fringe pattern is obtained by piecewise local correlation between the two images.²⁰ This fringe pattern is essentially a mapping of the change in the phase of the light (or equivalently the ray path length) from the two coherent sources to the scattering point on the object described by

$$\phi(r, \theta) = \frac{2\pi}{\lambda} (\vec{n}_1(r, \theta) - \vec{n}_2(r, \theta)) \cdot \vec{u}(r, \theta), \quad (9)$$

where $\vec{n}_1$ and $\vec{n}_2$ are unit vectors from the two sources to the surface point (r, θ) and $\vec{u}$ is the full three-dimensional stress-relief displacement. The wrapped corresponding fringe intensity I ranges from 0 (black) to 1 (white)²⁰ and is then given as

$$I(r, \theta) = (1 + \cos(\phi(r, \theta)))/2. \quad (10)$$

As such, a hypothetical fringe pattern expected from blind-hole drilling may be forward modeled by first determining the correct set of stress-relief displacements over the surface of the object, which is converted to a phase by eq (9) and then finally a fringe intensity by eq (10) that is subsequently used to produce an image. The forward modeled fringe pattern for the case in which $\delta = 2.38$, $r_0 = 0.2855$ cm, $E = 3.1$ GPa, and $\nu = 0.4$ (acrylic) shows a bow-tie pattern centered on the stress-relieving hole (Fig. 1(b)). This calculated fringe pattern matches qualitatively with that observed (Fig. 1(a)) of a stress-relief drilling of a hole with the above dimensions into a block of acrylic ($12.35 \times 12.35 \times 4.65$ cm³) subject to the measured uniaxial compression using the same experimental configuration as in earlier tests.¹⁰ It must be noted that the calculated fringe pattern in Fig. 1(b) is the direct result of inputting the actual values of the elastic moduli and the applied stress.

Conclusion

A series of parametric formulae describing the three-dimensional stress-relief displacement field induced by the drilling of a small stress-relieving hole has been constructed. The formulae are based on an extensive series of finite element models; descriptive coefficients are obtained by searching for the expressions that best fit the finite element results. These formulae, provided in dimensionless form, are capable of describing the stress-relief displacements for a wide range of hole depth and Poisson's ratio.

The greatest utility of these formulae will be to support stress determination using optical interferometric methods, many of which are sensitive, to some degree, to all three components of the displacement field. A second advantage is that, given the appropriate knowledge of the elastic moduli of the material and the hole dimensions, the formulae are well disposed to be employed in schemes which apply inversion theory to extract stress magnitudes from the fringe patterns.²² A contribution is forthcoming, which describes the analysis of laboratory tests such as shown in Fig. 1(a) using such a methodology and the formulae developed here. Finally, although this paper focuses solely on a uniform state of stress throughout the block, there is nothing that would preclude employing this methodology to more general states of stress that vary non-uniformly along the axis of the stress-relieving hole.

Acknowledgments

The computational support of L. Tober, M. Lazorek, and J. Haverstock is greatly appreciated. The experimental fringe pattern was produced by M. Diallo based on initial preparations by R.W. Hunt. This work was supported by a National Science and Engineering Research Council Strategic Grant to D.R. Schmitt.

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